

INPUT

BASES

MIX

PREDICTION

Graph
input $X, \hat{A}$

features, adjacency

Low-pass basis

$$L_k = \hat{A}^k X$$

Residual basis

$$H_k = L_{k-1} - L_k$$

Node variation

$$r_i = \|x_i - \hat{A}x_i\|$$

Low mix

$$Z_L = \sum_k \alpha_k \phi(L_k)$$

High mix

$$Z_H = \sum_k \beta_k \psi(H_k)$$

Gate

$$g_i = \sigma(\text{MLP}(r_i))$$

Gated fusion

$$Z_i = (1 - g_i)Z_{L,i} + g_i Z_{H,i}$$

 $\hat{y}_i$